

Relational Algebra

Spring 2024

- b)
- classroom** (building, room number, capacity)
 - department** (dept_name, building, budget)
 - course** (course_id, title, dept_name, credits)
 - instructor** (ID, name, dept_name, salary)
 - section** (course_id, secid, semester, year, building, room number, time slot id)

teaches (ID, course_id, sec_id, semester, year)
student (ID, name, dept_name, tot_cred)
takes (ID, course_id, sec_id, semester, year, grade)

Consider the relational database of a university database management system, where the primary keys are underlined. Write queries in relational algebra to answer the following questions -

1. Find the ID, name and salary of each instructor in the Physics department or those having salary greater than \$40,000

$\pi_{ID, name, salary} (\sigma_{(dept_name = 'Physics' \vee salary > 40000)} (instructor))$

2. Find the ID and name of each instructor in a department located in the building 'Watson'

$\pi_{ID, name} (\sigma_{(building = 'Watson')} (instructor \bowtie department))$

3. Find all the courses taught in the Fall 2023 semester but not in Spring 2024 semester.

$r1 \leftarrow \pi_{course_id} (\sigma_{semester = 'Fall' \wedge year = 2023} (section))$

$r2 \leftarrow \pi_{course_id} (\sigma_{semester = 'Spring' \wedge year = 2024} (section))$

$r1 - r2$

4. Find the ID and name of those instructors who earn more than the instructors in the Physics department.

```
r1 <-  $\pi$  salary ( $\sigma$  dept_name = 'Physics' (instructor))  
r2 <-  $\pi$  ID, name ( $\sigma$  salary > (avg(r1)) (instructor))
```

or,

1. Find all instructor whose salary is above average for their department

```
r1 <-  $\pi$  dept_name, avg(salary) (instructor)  
r2 <-  $\pi$  ID, name, dept_name ( $\sigma$  salary > avg_salary (instructor  $\bowtie$  r1))
```

2. Find all courses taught in both the Fall 2022 semester and in the Spring 2022 semester

```
r1 <-  $\pi$  course_id ( $\sigma$  semester = 'Fall'  $\wedge$  year = 2022 (section))  
r2 <-  $\pi$  course_id ( $\sigma$  semester = 'Spring'  $\wedge$  year = 2022 (section))  
r3 <- r1  $\cap$  r2
```

3. Find all students who have taken all courses taught by instructor named 'John Doe'

```
r1 <-  $\pi$  course_id ( $\sigma$  name = 'John Doe' (instructor)  $\bowtie$  teaches)  
r2 <-  $\pi$  student_id (takes  $\bowtie$  r1)  
r3 <- r2  $\div$  r1
```

4. List all departments along with number of students in each department.

$r1 \leftarrow \pi_{dept_name, student_id} (student)$

$r2 \leftarrow \pi_{dept_name} (r1)$

Autumn 2023

Customer (customer_id, customer_name, address, city, state, phone_number)
Store (store_id, store_name, store_address, store_phone)
Food (food_id, store_id, food_name, food_image, food_category, food_price)
Order (order_id, customer_id, store_id, food_id, shipment_address, payment_method, quantity, total_price, order_date)

Consider the relational database of an online food app, where the primary keys are underlined. Write **queries in relational algebra** to answer the following questions -

1. Find the names of customers who has ordered between 1000 Tk and 2000 Tk.

$r1 \leftarrow \sigma_{total_price \geq 1000 \wedge total_price \leq 2000} (Order)$

$r2 \leftarrow \pi_{customer_name} (Customer \bowtie r1)$

2. Find the food items with their prices from a specific store. (store_name = "Well Food")

$r1 \leftarrow \sigma_{store_name = 'Well Food'} (Store)$

$r2 \leftarrow \pi_{food_name, food_price} (Food \bowtie r1)$

3. Find the total number of food items available in each store.

$\pi_{\text{store_id, food_id}} (\text{Food} \bowtie \text{Store})$

$r3 \leftarrow \gamma_{\text{store_id, count(food_id)}} (r2)$

4. Find the order details of those customers who lives in the same city where the customer "X" lives.

$r1 \leftarrow \pi_{\text{city}} (\sigma_{\text{customer_name} = 'X'} (\text{Customer}))$

$r2 \leftarrow \sigma_{\text{city} = r1.\text{city}} (\text{Customer} \bowtie r1)$

$r3 \leftarrow r2 \bowtie \text{Order}$

Autumn 2022

a. Consider the Following Relational Database of *Pure Islamic Bank*:

branch (branch_name, branch_city, assets)

customer (customer_name, customer_street, customer_city)

account (account_number, branch_name, balance)

loan (loan_number, branch_name, amount)

depositor (customer_name, account_number)

borrower (customer_name, loan_number)

Write Relational Algebra Expression (Any three)

(i) Find all customers who have a loan, an account, or both.

$r1 \leftarrow \pi_{\text{customer_name}} (\text{depositor})$

$r2 \leftarrow \pi_{\text{customer_name}} (\text{borrower})$

$r3 \leftarrow r1 \cup r2$

(ii) Find all customers who have a loan at the *Kumira* branch.

$\pi_{\text{customer_name}} (\sigma_{\text{branch_name} = \text{'Kumira'}} (\text{borrower} \bowtie \text{loan}))$

(iii) Find the names of all customers who have a loan in *GEC* branch but do not have an account in the bank.

$r1 \leftarrow \pi_{\text{customer_name}} (\sigma_{\text{branch_name} = \text{'GEC'}} (\text{borrower} \bowtie \text{loan}))$

$r2 \leftarrow \pi_{\text{customer_name}} (\text{depositor})$

$r3 \leftarrow r1 - r2$

(iv) Find the account numbers of the Millionaires of the bank.

$\pi_{\text{account_number}} (\sigma_{\text{balance} \geq 1000000} (\text{account}))$